Byegerm1.0 Reproduction Guide

Fixed dual-tube instrument: fabrication, assembly and system commissioning

1 Technical field

Byegerm1.0 combines temperature control, fluorescence imaging, image analysis and wireless communication in a portable enclosure. This manual describes the components of each module and explains how to replicate the instrument, from structural fabrication and electrical connections to optical setup and system commissioning.

2 Background

Portable testing requires stable reaction conditions and consistent imaging conditions. Designed for home use, byegerm1.0 has two fixed tube positions and a mobile app that makes temperatures, images and test records easier to view, reducing reliance on a computer during operation.

3 System design

The instrument measures approximately 7.5 × 14 × 11 cm and is compact enough for one person to carry. Its internal components are arranged in five modules: optics, temperature control, power management, AI data processing and Wi-Fi communication.

Module	Main components and functions
Optics	A 1 W blue LED sits below each tube slot. A camera, lens and optical filter components are positioned in front to capture fluorescence images.
Temperature control	Two ceramic heater cores with a combined power of 6 W work with temperature feedback, PID control and a DC fan to provide heating and cooling.
Power management	A lithium battery, balancing and protection components, and charging management components supply power to the modules. LEDs indicate the remaining battery charge.
AI data processing	An Orange Pi 5 Max with an RK3588 processor runs YOLOv8n and OpenCV to locate tube positions and analyze images.
Wi-Fi communication	A TCP/IP connection to a socket on the phone carries images, temperatures and analysis results.

4 Figures

Figure 1 shows the instrument structure and computing board. Figures 2 and 3 show photographs of the components and assembly. Figure 4 covers the interfaces, temperature sensing and lighting circuits. Figure 5 compares camera mounting distances, and Figure 6 shows the app's image display. The figures appear alongside the relevant steps.

Before fabrication, prepare the structural parts, circuit boards, wiring harnesses and fasteners, along with screwdrivers, a multimeter, soldering tools and tools for preparing the wiring harnesses. Perform a trial assembly without power, then connect and test each module.

5 Implementation

5.1 Structural fabrication and module assembly

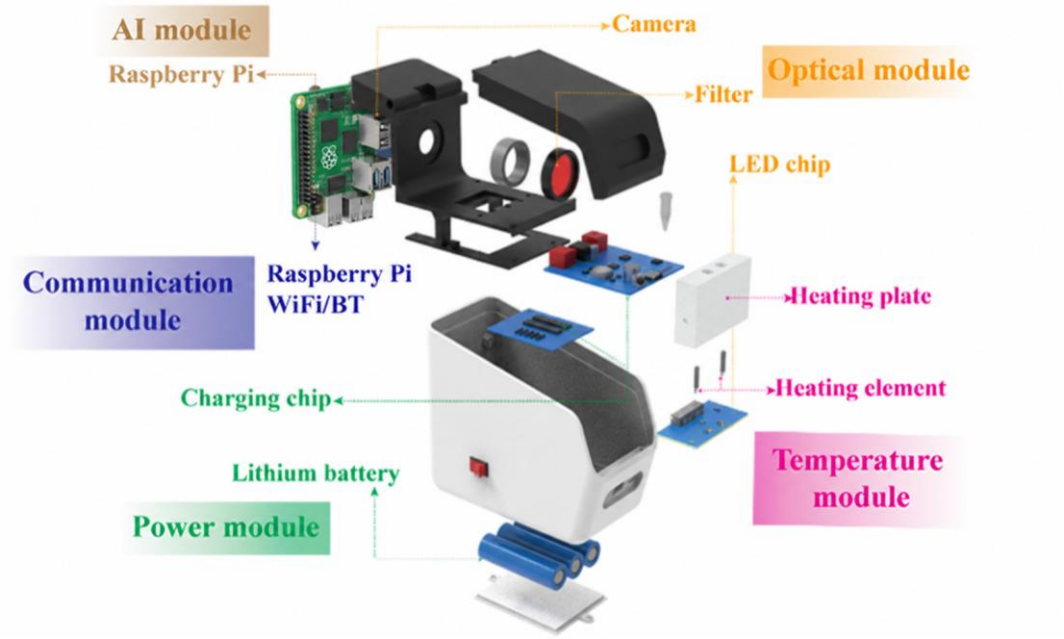


Figure 1 Exploded view of the instrument modules

1 Fabricate the structural parts. Model the enclosure, brackets and working chamber in SolidWorks 2024. Check the hole positions and interfaces, and confirm that the cover opens and closes properly. Export and slice the models, then 3D-print the parts. Remove the supports and perform a trial assembly before installing the internal components.

2 Assemble the temperature-control components. Secure the dual-tube metal block to its support. Fit the two ceramic heater cores closely around the tube-slot cavity, and secure the sensor at the designated temperature measurement point. Position the fan on the right and install the insulation as shown in Figure 1, keeping the airflow path clear.

3 Install the computing and power modules. Secure the computing board, battery and power management components separately, leaving room for cooling and connector access. Fit the camera bracket loosely, then tighten it after setting the imaging position in Section 5.3.

Physical components and assembly reference

Figure 2 shows the temperature-control and optical components and their locations inside the enclosure, as described on the preceding page. Figure 3 shows the reaction-area assembly being secured to the enclosure. Refer to Figure 1 to check how the parts fit together.

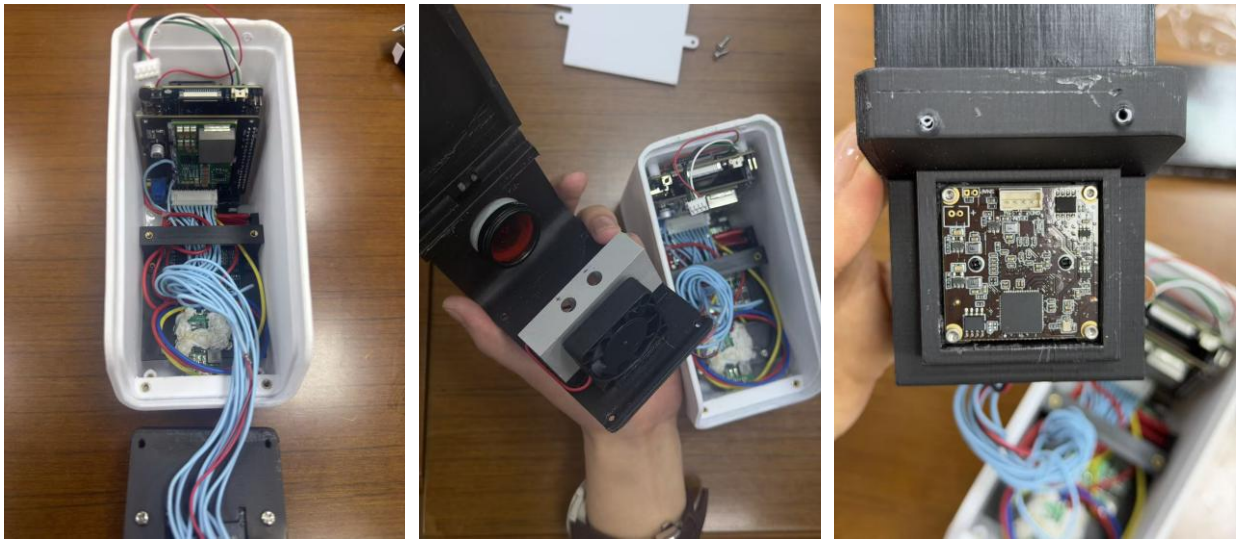


Figure 2 Photographs of the internal components of byegerm1.0. Left: circuit boards and wiring layout. Center: dual-tube metal block, fan and optical filter assembly. Right: circuit board on the back of the camera and its bracket.



Figure 3 Photograph of the byegerm1.0 reaction-area assembly, showing how the support plate, metal support block, fan and enclosure are secured together.

5.2 Electrical connections and power supply

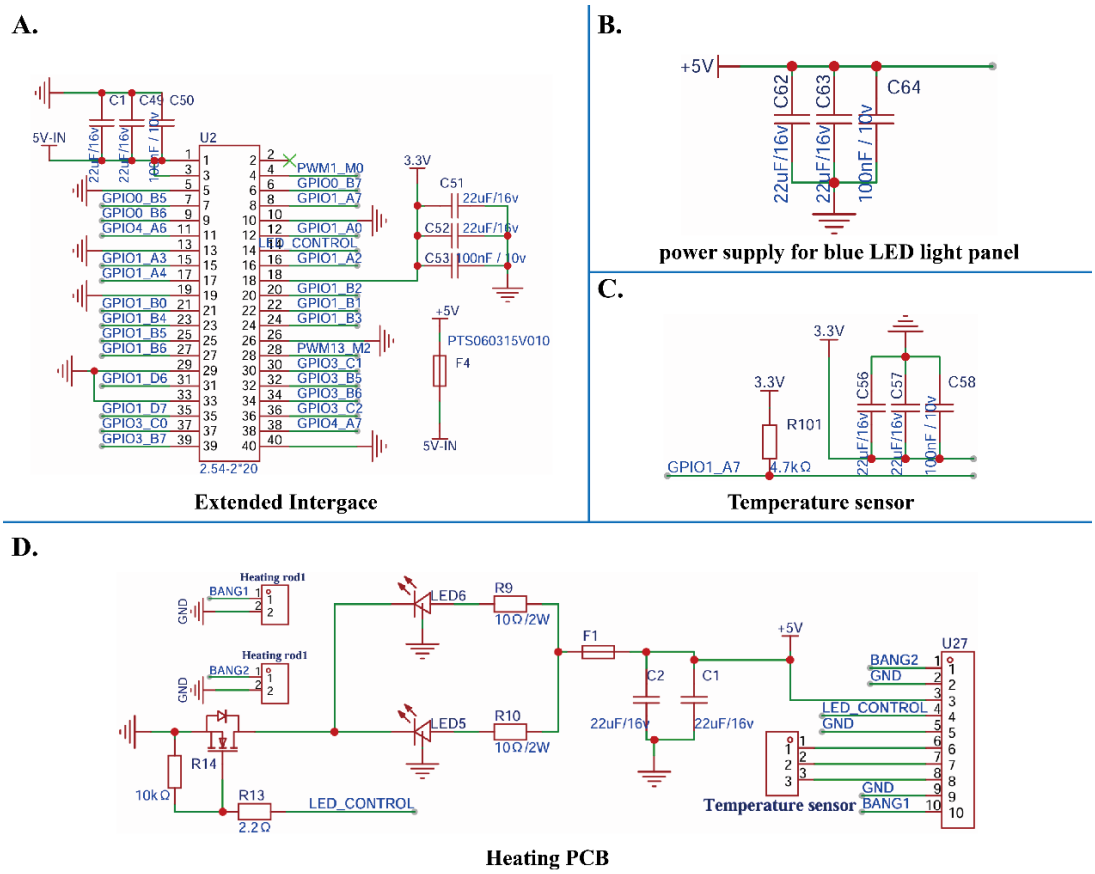


Figure 4 Expansion interfaces and working-chamber circuits. A: expansion interfaces. B: LED power supply. C: temperature-sensing interfaces. D: heating and lighting connections.

The functional modules connect through wiring harnesses with terminal connectors. Use the net names in Figure 4 to check the power, signal and ground connections, then confirm the orientation and pin 1 position of each connector. Route the wiring neatly to reduce assembly errors and simplify later troubleshooting and maintenance.

1 Connect the working chamber. Connect the heaters, temperature sensors, fan and LEDs to their corresponding driver or expansion interfaces. Power the heating elements and LED boards through the driver circuits, not directly from the computing board's GPIO pins. Keep the harnesses clear of the tube openings, fan and camera field of view.

2 Connect the camera and power supply. Connect the camera to the computing board via USB. Keep the battery disconnected while checking the power assembly's inputs, outputs and common ground. Use a multimeter to confirm that the polarity and voltage match the ratings of the boards and loads.

3 Power up the components in sequence. Start the computing board and read the raw images and temperatures first. Then enable the lighting, fan and heating in that order. Confirm that each control operates the intended component and that the power supply is stable before securing the harnesses.

5.3 Optical installation and image acquisition settings

Install a blue LED below each of the two tube slots. Position a USB industrial camera with an ultra-macro lens directly in front of the slots. Fit suitable optical filter components in front of the lens to suppress excitation light and capture fluorescence.

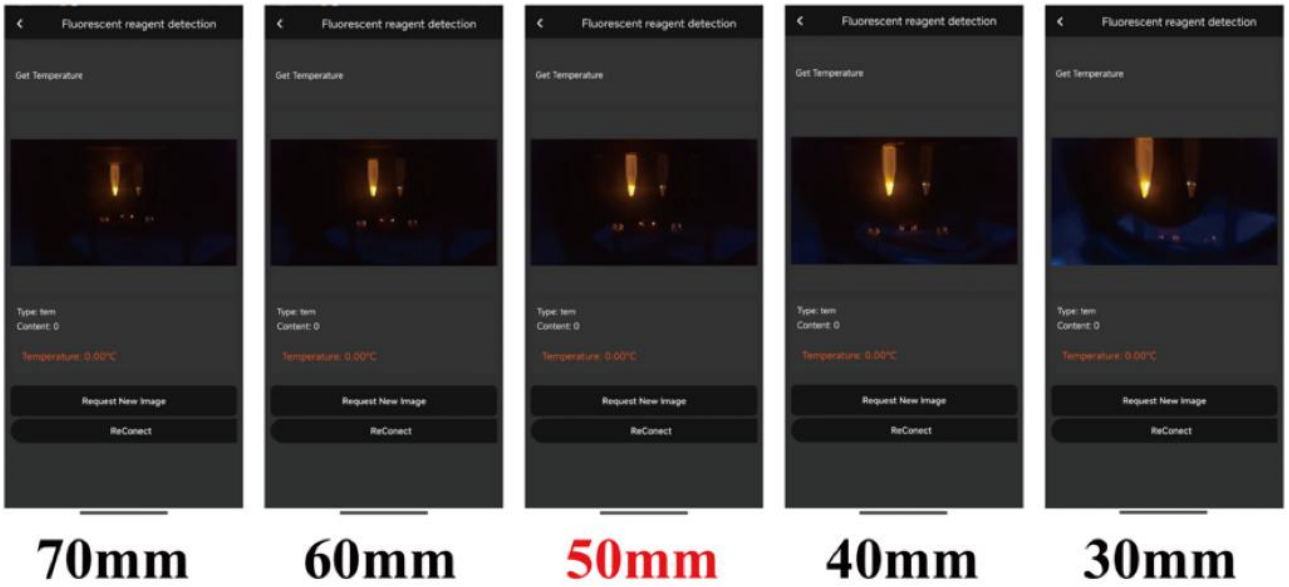


Figure 5 Comparison of images captured at different distances between the camera and tube positions

1 Adjust the imaging position. Use the 50 mm setup in Figure 5 as a reference. Adjust the bracket and focus until the reading regions of both tubes are fully visible and sharp. Close the cover and check for light leakage, then tighten the bracket once the position is set.

2 Set consistent acquisition conditions. Use the same reference object to adjust exposure, gain and white balance, checking for reflections and overexposure. Save images of the empty chamber and images with the lights off and on. Capture a continuous sequence to confirm image stability.

5.4 Image processing and communication setup

The Raspberry Pi uses a high-performance 2.4 GHz 64-bit ARM-based SoC and runs the trained YOLOv8n model and an OpenCV image-processing program. The instrument uses the trained YOLOv8n model to locate the reaction tubes, then uses OpenCV to extract and analyze the corresponding image regions before sending the results to the app.

5.5 App connection and system commissioning

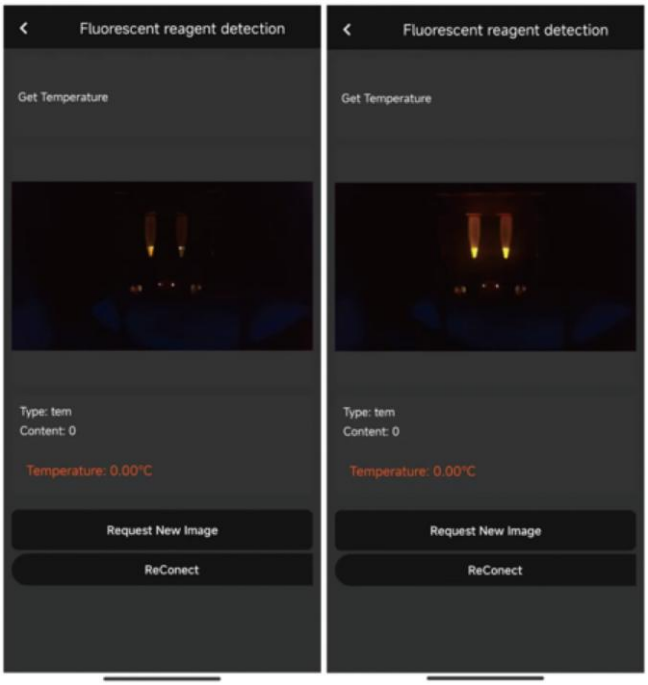


Figure 6 Example of the app displaying received images and device information

1 Establish the connection. Start the instrument and connect it and the phone to the same local network. Enter the instrument's Server IP and Port in the app, select Connect to Server, and confirm that new data arrive.

2 Configure settings and acquire images. Check the temperature and time settings under Settings, then send an image request. Verify the acquisition time, left and right tube positions, and saved records. Test continuous acquisition and reconnection after a dropped connection.

3 Test temperature control and imaging together. At the fixed measurement point, compare the software reading with a reference thermometer. Check the temperature response during warm-up, steady operation and after stopping. Capture images again while temperature control is running to confirm stable lighting, power and imaging.

When commissioning is complete, save the raw images, temperature records and instrument configuration as a reference for maintenance. Restart the instrument and check that the configuration is restored and that communication and acquisition work. After replacing a camera, sensor or heating component, recommission the affected module.

Symptom	Items to check
Instrument will not start	Input power, switch, protection components and power to the computing board
Abnormal temperature	Sensor wiring, mounting position and software input channel
Image too dark or overexposed	Lighting, optical filter components, focus and exposure settings
No new data in the app	Local network, address and port, communication service and response time